mass and its circumference. All the masses were less than 1.4 Suns, as he had firmly expected. Moreover, when he plotted the stars' masses and circumferences on a diagram and "connected the dots," he obtained a single, smooth curve (right half of Figure 4.3; see also Box 4.2), and the measured masses and circumferences of Sirius B and other known white dwarfs agreed with that curve moderately well. (With improved, modern astronomical observations, the fit has become much better; note the new, 1990 values of the mass and circumference of Sirius B in Figure 4.3.) Proud of his results and anticipating that the world's astronomers would finally accept his claim that white dwarfs cannot be heavier than 1.4 Suns, Chandrasekhar was very happy.

Especially gratifying would be the opportunity to present these results to the Royal Astronomical Society in London. Chandrasekhar was scheduled for a presentation on Friday, 11 January 1935. Protocol dictated that the details of the meeting's program be kept secret until the meeting started, but Miss Kay Williams, the assistant secretary of the Society and a friend of Chandrasekhar's, was in the habit of sending him programs secretly in advance. On Thursday evening when the program arrived in the mail, he was surprised to discover that immediately following his own talk there would be a talk by Eddington on the subject of "Relativistic Degeneracy." Chandrasekhar was a little annoyed. For the past few months Eddington had been coming to see him at least once a week about his work and had been reading drafts of the articles he was writing, but never once had Eddington mentioned doing any research of his own on the same subject!

Box 4.2

An Explanation of the Masses and Circumferences of White-Dwarf Stars

To understand qualitatively why white dwarfs have the masses and circumferences shown in Figure 4.3, examine the drawing below. It shows the average pressure and gravity inside a white dwarf (plotted upward) as functions of the star's circumference (plotted rightward) or density (plotted